

Mode-Seeking Meets Autoregression: A Training-Free Factorial Diagnosis of Diversity Collapse in Few-Step Video Generators

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Abstract

001 Few-step distillation makes video diffusion models fast, and
002 a growing literature reports that it also makes them worse:
003 motion collapses, diversity shrinks, colours saturate. Ev-
004 ery such report comes from a paper proposing a fix, each
005 with its own metric, so the losses cannot be compared, and
006 none of them separates the effects of fewer steps, rewrit-
007 ten weights and causal (autoregressive) architecture, which
008 real streaming students combine. We disentangle them with
009 a 2×2 factorial design built entirely from public check-
010 points of one model family (Wan2.1-1.3B): {bidirectional,
011 causal} $\times$ {50-step, few-step distilled}, replicated in two in-
012 dependent labs, at three teacher guidance scales, across
013 20 student recipes and 5,405 videos, with a five-axis, non-
014 aggregated evaluation protocol calibrated against blind
015 pairwise human judgements. We find that the loss of
016 cross-seed diversity is an interaction: causalisation alone
017 and bidirectional DMD alone cost little, while their com-
018 bination collapses diversity (interaction -0.063 , 95% CI
019 $[-0.092, -0.036]$, sign-stable across guidance scales and
020 feature spaces). The interaction is driven by the distilla-
021 tion objective, not the architecture: it vanishes under con-
022 sistency distillation and halves under adversarial training,
023 and more than half of it is already incurred at the ODE-
024 regression initialisation stage that precedes DMD. It is fully
025 present in the first generated chunk and then merely prop-
026 agated by the KV cache; chunk size does not modulate
027 it. Taxes on other axes have opposite signs—causalisation
028 costs temporal coherence, which distillation refunds—so no
029 scalar can rank these models. Finally, four frame-level
030 image-quality metrics rank distilled students above their
031 teacher while humans rank the teacher far above all stu-
032 dents: the “quality improves after distillation” numbers in
033 the literature are a metric artefact. Corpus, per-video met-
034 rics and scripts are released.

1. Introduction

Streaming and interactive video generation runs on few-
step, causal students distilled from bidirectional multi-step
teachers: CausVid [20], Self-Forcing [8], Causal For-
cing [24], Causal-rCM [22]. The cost of this speed-up is
visible to anyone who samples the same prompt with dif-
ferent seeds—the student returns the same shot again and
again (Fig. 1)—and it has been documented repeatedly:
dynamic degree dropping from 0.80 to 0.00 [4], diversity
collapse [3, 5], temporal collapse and oversaturation [1].
Each of these papers proposes a remedy and reports the
degradation that its remedy repairs, with its own metric
and its own baseline. Three things are therefore still un-
known. (i) *Whose fault is it?* A causal DMD student dif-
fers from its teacher in three ways at once—fewer sampling
steps, distilled weights, and a causal architecture with a KV
cache—and no comparison in the literature includes the cell
that isolates the third: a causal model that is *not* distilled.
(ii) *Which recipe?* DMD, SiD, consistency, adversarial and
ODE-regression objectives are all in use, with and without
self-rollout training, at different chunk sizes; their losses
have never been measured under one protocol. (iii) *Which
metric?* Distillation papers routinely report that imaging-
quality scores go *up* while motion goes down [4]; whether
that reflects perception has not been tested.

We answer all three without training anything. The
Wan2.1-1.3B family is unique in that public checkpoints ex-
ist for every cell of a 2×2 factorial design—bidirectional or
causal architecture, 50-step or few-step distilled sampling—
from two independent labs (CausVid/Causal-Forcing and
NVIDIA Causal-rCM), together with twelve further stu-
dents that vary the objective (DMD, SiD, GAN, dis-
crete/continuous consistency), the training stage (ODE-
regression init, consistency init, DMD refinement), the ex-
posure regime (teacher forcing vs. self-rollout) and the
chunk size (1 or 3 latent frames). We generate every con-
figuration on the same 30 prompts and the same initial
noise, measure five capability axes with one primary met-

073 ric each (mode coverage, cross-seed diversity, object motion,
074 prompt adherence, temporal coherence), never aggregate
075 across axes, and calibrate the axes that matter against
076 90 blind pairwise human judgements.

077 **Findings.** (1) **The diversity tax is an interaction.**
078 Causalising the teacher costs almost no diversity (-0.006),
079 bidirectional DMD costs little (-0.017), but the two
080 together cost -0.086 ; the interaction term is -0.063
081 [$-0.092, -0.036$] and keeps its sign when the teacher’s
082 guidance scale is changed from 3 to 7.5 and when the
083 feature space is changed (§4.1). (2) **The interaction be-**
084 **longs to the objective, not the architecture.** With a
085 consistency objective the same causal architecture shows
086 *no* negative interaction (and a positive one on cover-
087 age, $+0.48$ [$0.28, 0.69$]); adding self-rollout DMD on top
088 of the consistency initialisation brings it back (-0.035
089 [$-0.063, -0.008$]); SiD behaves like DMD; GAN col-
090 lapses about half as much (§4.2). (3) **Most of the loss pre-**
091 **cedes DMD.** Along the public training ladders, the ODE-
092 regression initialisation already removes more than half of
093 the diversity that the final DMD student lacks; consistency
094 initialisation does not (§4.3). (4) **It happens in the first**
095 **chunk.** Per-frame diversity of causal DMD students is fully
096 depressed at frame 0 and flat afterwards; undistilled causal
097 models instead *gain* diversity along the video through au-
098 toregressive drift, which is where their temporal tax comes
099 from. Chunk size (1 vs. 3 latent frames) does not modulate
100 the collapse—a mechanism prediction we made and that
101 failed (§4.3). (5) **Taxes have opposite signs across axes.**
102 Causalisation costs temporal coherence and distillation re-
103 funds it (positive interaction in every configuration); prompt
104 adherence is untaxed; camera motion is taxed two to three
105 times more than object motion; oversaturation appears with
106 causal fine-tuning, not with DMD (§4.4). (6) **Frame-level**
107 **quality metrics are inverted on distilled video.** MUSIQ,
108 CLIP-IQA, MANIQA and TOPIQ all rank DMD students
109 above the teacher; a human ranks the teacher above every
110 student with a 35.5/36 win rate, and two of the metrics agree
111 with the human significantly *below* chance (§4.5).

112 2. Related work

113 **Few-step and causal distillation.** DMD [18, 19] matches
114 the student’s output distribution to the teacher’s score
115 through a fake-score network and is the workhorse of video
116 distillation (FastWan [12], CausVid [20]). CausVid converts
117 the bidirectional Wan2.1 into a block-causal autoregressive
118 model with a KV cache; Self-Forcing [8] trains on the stu-
119 dent’s own rollouts to remove exposure bias; Causal For-
120 cing [24] adds a teacher-forced causal diffusion stage and an
121 asymmetric DMD; Causal-rCM [21, 22] replaces the ini-
122 tialisation with teacher-forced consistency distillation be-
123 fore self-forcing DMD. SiD [23] and adversarial variants

are released alongside Self-Forcing. We use all of these as
they were published; we train nothing.

Reported degradations. DynaForcing [4] documents the
dynamic-degree collapse of Self-Forcing under extended
training; Adaptive Video Distillation [1] attributes over-
saturation and temporal collapse to distribution-matching
losses; Data-Forcing [3] and Uncertainty DMD [5] at-
tribute diversity collapse to DMD’s mode seeking, the lat-
ter proposing that a collapsed first chunk is propagated by
the deterministic cache; CoDMD [2] shows the sign of the
degradation flips with model scale. All are method papers.
We confirm the mechanism of [5] with a factorial design
that isolates it, and add that it is objective-specific, mostly
incurred before DMD, and invisible to standard quality met-
rics.

Evaluation. VBench [9] aggregates sixteen dimensions;
its dynamic-degree and smoothness dimensions are anti-
correlated by construction, which we quantify. Preci-
sion/recall in feature space [11, 13] separate fidelity from
coverage; we use them on VideoMAE [15] and DI-
NOv2 [14] features. No-reference IQA (MUSIQ [10],
CLIP-IQA [16], MANIQA [17], TOPIQ [6]) is used as
“imaging quality” in video benchmarks; we test it against
human judgement on distilled video.

513 3. Protocol

Configurations. All models share the Wan2.1-T2V-1.3B
backbone, the same VAE, the same cached UMT5-XXL
prompt embeddings, the same attention kernel and the same
generation setting (81 frames, 480×832 , 16 fps, identi-
cal initial noise per seed). Table 1 lists the cells. The
teacher (T-FULL) uses Wan’s default sampler (UniPC, 50
steps, shift 5, CFG 5). The undistilled causal cells (AR-
DIFF) are the teacher-forced causal diffusion checkpoints
released by Causal-Forcing (chunk of 3 latent frames)
and by Causal-rCM (TF-Diffusion), sampled with the
same UniPC/CFG setting through their KV-cache pipeline.
Bidirectional distilled cells are CausVid’s bidirectional
DMD checkpoint and NVIDIA’s rCM; causal distilled cells
are CausVid’s autoregressive DMD, Self-Forcing (DMD,
SiD, SiD-v2, GAN), Causal-Forcing (DMD, consistency,
ODE-init, frame-wise), and Causal-rCM (TF-dCM, TF-
dCM+SF-DMD, both at chunk 3 and chunk 1). Each
student is sampled with its authors’ own schedule. Fast-
Wan [12] is included as a secondary bidirectional DMD
cell (it was trained with sparse attention; we run it densely).
Two sets of controls address confounds: T-FEW (the teacher
at 4 steps) isolates *fewer steps*; T-CFG{1, 3, 7.5} and AR-
DIFF-cfg{3, 7.5} trace the guidance scale, because distilled
students absorb guidance and DMD targets the teacher at a

real-score guidance of 3–3.5.

Prompts and seeds. 30 prompts fixed before any generation: 15 VBench sentences covering static, slow and dynamic content, and 15 written to hit the regimes reported as distillation-sensitive (fast motion, multi-object interaction, rare compositions, object-count conservation, physical events). Eight seeds per prompt; seeds map to identical initial latent noise in every cell, so all cell comparisons are paired.

Five axes, one primary metric each. **A1 coverage:** k -NN manifold recall of the teacher’s samples by the student’s ($k=3$) [11], pooled over prompts, VideoMAE-base features (DINOv2-L frame features as a second space); precision is always reported alongside so that “garbage is diverse” cannot pass. **A2 diversity:** mean pairwise cosine distance among the 8 seeds of a prompt in the same feature spaces. **A3 motion:** RAFT flow decomposed by robust affine fitting into a global (camera) and a residual (object) field; the residual magnitude is primary, the global one reported. **A4 adherence:** BLIP-2 yes/no VQA on three pre-registered questions per prompt (CLIP-T alongside). **A5 temporal coherence:** RAFT backward-warp error (lower is better). We never combine axes into a score.

Statistics. Every contrast is paired over the 30 prompts (the same seeds and noise appear in every cell): prompt-level bootstrap CIs for A2–A5, leave-one-prompt-out jack-knife for the set-level A1. The 2×2 interaction is $I = (D_{\text{causal}} - D_{\text{bi}})_{\text{distilled}} - (D_{\text{causal}} - D_{\text{bi}})_{50\text{-step}}$ computed per prompt and bootstrapped. Rank structure across cells is summarised by the eigenvalue spectrum of the axis-correlation matrix and by counting significant rank reversals (two cells ordered oppositely on two axes, both $|z| > 1.96$).

Human calibration. One rater (an author; a limitation) judged 90 blind pairwise comparisons: for a prompt, four seeds of cell A vs. four seeds of cell B , autoplaying side by side, answering which side is more diverse across seeds, which has more natural motion, and which has better image quality. Six cells, a balanced round-robin design (each of the 15 pairs six times, left/right randomised); Bradley–Terry scores with ties as half wins.

4. Results

4.1. The diversity tax is an interaction

Table 2 gives the primary metrics for the key cells; Fig. 2 the factorial decomposition. Causalising the teacher without distilling it leaves cross-seed diversity essentially intact ($0.161 \rightarrow 0.155$) and costs a quarter of the coverage; bidirectional DMD costs a little diversity ($\rightarrow 0.144$)



Figure 1. One prompt (“three children chasing each other around a fountain”), four seeds, middle frame. Neither causalising the teacher (row 2) nor bidirectional DMD (row 3) destroys cross-seed variety; the causal DMD student (row 4) returns the same composition for every seed.

and half the coverage; the causal DMD student loses more than either sum: diversity 0.075, recall 0.03. The prompt-paired interaction is -0.063 [$-0.092, -0.036$] ($P(I \geq 0) < 10^{-4}$), larger in magnitude than both main effects together. It replicates in DINOv2 feature space (-0.065 [$-0.115, -0.017$]) and is sign-stable when the two 50-step cells are re-sampled at CFG 3 (-0.036 [$-0.064, -0.007$]) or CFG 7.5 (-0.047 [$-0.076, -0.018$]), which brackets the guidance the students actually absorbed. The interaction is visible to the eye (Fig. 1) and to the rater: the causal DMD student won 3.5 of 25 diversity comparisons.

Two alternative explanations are excluded. *Fewer steps:* the 4-step teacher is worse than every student on coverage (recall 0.35, precision 0.50) and adherence, but *more* diverse than the teacher (0.169)—few-step sampling produces blurry, varied garbage, not a collapsed mode; distillation relative to it is a refund. *High effective guidance:* the teacher’s diversity falls from 0.198 (CFG 3) to 0.161 (5) and saturates at 0.157 (7.5); bidirectional DMD students (0.144–0.154) sit near the teacher’s high-guidance end, but causal DMD students (0.075–0.134) lie below any guidance setting of the teacher (Fig. 6, right).

4.2. The interaction belongs to the objective

Replacing the causal DMD cell by other students keeps the two 50-step cells fixed and changes only the objective (Fig. 3, Table 2). With NVIDIA’s teacher-forced *consistency*-distilled causal student the interaction on diversity is $+0.011$ [$-0.018, +0.040$], on DINOv2 diversity $+0.166$ [$0.116, 0.220$] and on coverage $+0.48$ [$0.28, 0.69$]: a mode-covering objective on a causal architecture *recovers* coverage rather than losing it. Refining that same student with self-rollout DMD brings the negative interaction

Table 1. Cells of the factorial design (all Wan2.1-1.3B, 81 frames, 480×832). TF = teacher-forced context during training, SR = self-rollout. n = videos with 8 seeds unless noted.

cell	architecture	objective / stage	source
T-FULL	bidirectional, 50 steps	—	Wan2.1
AR-DIFF	causal, 50 steps	TF diffusion	Causal-Forcing
AR-DIFF-rcm	causal, 50 steps	TF diffusion	Causal-rCM
S-BI-causvid	bidirectional, 3 steps	DMD	CausVid
S-BI-rcm	bidirectional, 4 steps	consistency (rCM)	NVIDIA
S-CAUSAL-causvid	causal, 3 steps	DMD, TF	CausVid
S-sf / -sid / -sid2 / -gan	causal, 4 steps	DMD / SiD / GAN, SR	Self-Forcing
S-cf / -cd / -ode / -fw	causal, 4 steps	DMD(SR) / CM / ODE-init / chunk 1	Causal-Forcing
S-rcm-tfdcm (-fw)	causal, 4 steps	consistency, TF (chunk 1)	Causal-rCM
S-rcm-sfdmd (-fw)	causal, 4 steps	CM init + SR-DMD (chunk 1)	Causal-rCM
S-causvid-ode	causal, 3 steps	ODE-regression init	CausVid
T-few, T-cfgc, AR-diff-cfgc	controls	4 steps; CFG c ($n=4$ seeds)	—

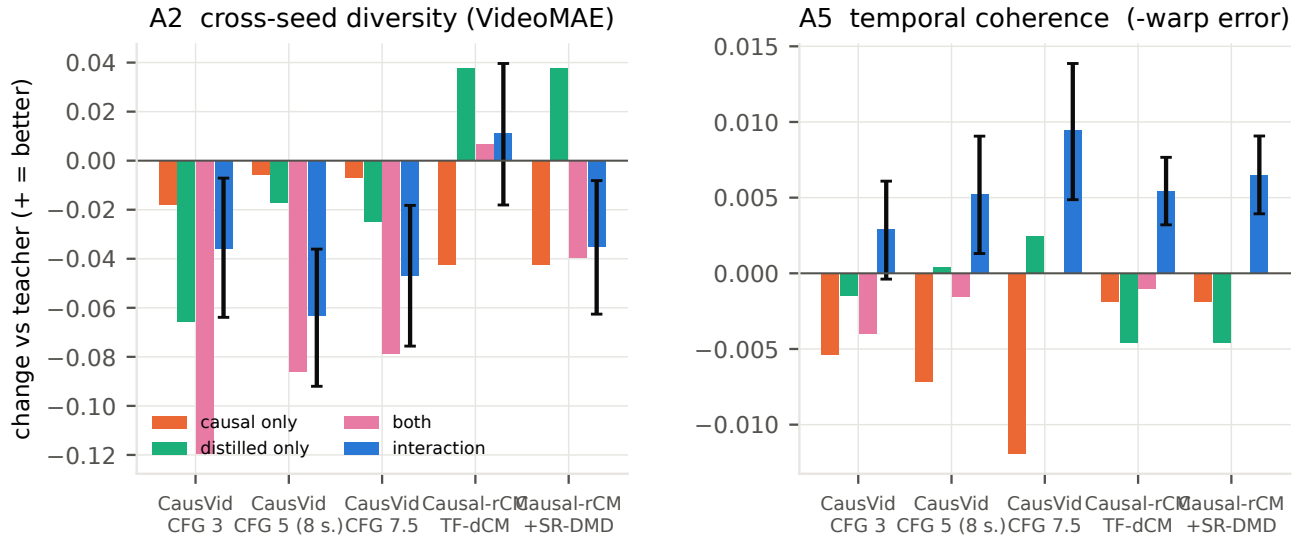


Figure 2. 2×2 main effects and interaction for diversity (left) and temporal coherence (right), each bar the change from the teacher, error bars the 95% prompt-paired bootstrap CI of the interaction. Left three groups: CausVid cells with the teacher at CFG 3/5/7.5. Right two: the NVIDIA Causal-rCM cells with a consistency-distilled causal student (no interaction) and the same student after self-rollout DMD (interaction returns). Temporal coherence shows the opposite sign everywhere: causalisation costs it, distillation refunds it.

back (-0.035 [-0.063 , -0.008]). Within the Self-Forcing family, where architecture, data and self-rollout training are held fixed, SiD and SiD-v2 are indistinguishable from DMD on diversity (differences ≤ 0.01 , CIs cross zero) while the adversarial student is significantly more diverse ($+0.034$ [0.022 , 0.047], recall 0.47 vs. 0.32). The two consistency cells from two labs and the adversarial cell also show what those objectives pay instead: object motion of 1.35–2.4 px/frame against the teacher’s 3.7, and precision of 0.53–0.56. Self-rollout training, the fix proposed by Self-Forcing, halves the interaction ($-0.063 \rightarrow -0.035$) but does not remove it; its entire benefit over CausVid is in the

interaction term—the main effects are unchanged.

4.3. Where and when the loss is incurred

Stage. The public ladders let us place the loss in training time. In CausVid, the ODE-regression initialisation—a causal model regressed onto teacher ODE trajectories, no DMD yet—already has diversity 0.094 against the teacher’s 0.161; DMD refinement takes it to 0.075 (paired difference $+0.019$ [0.006 , 0.033]; DINOv2 $+0.149$ [0.109 , 0.186]). More than half of the final deficit is incurred before any distribution matching: regressing each noise onto a single teacher endpoint is itself mode-seeking. In Causal-Forcing the ladder is monotone in both directions—consistency

Table 2. Primary metrics for the key cells (30 prompts $\times$ 8 seeds). Recall/precision are k -NN manifold metrics against the teacher in VideoMAE space; diversity is the pairwise cosine distance over seeds; motion in px/frame; VQA is BLIP-2 $P(\text{yes})$; warp error lower is better; colourfulness [7]; MUSIQ is VBench’s imaging-quality metric; human BT scores from §4.5.

cell	A1 coverage		A2 diversity		A3 motion		A4	A5	colour	MUSIQ	human div / qual
	recall	prec.	VideoMAE	DINOv2	object	camera	VQA	warp			
T-FULL	0.97	0.88	0.161	0.350	3.69	5.71	0.678	0.016	51	68.2	+1.1 / +3.7
AR-DIFF(Causal-Forcing)	0.61	0.62	0.155	0.303	3.44	4.83	0.672	0.023	77	65.8	+0.4 / -0.6
AR-DIFF-rcm (NVIDIA)	0.39	0.75	0.119	0.268	3.02	4.10	0.666	0.018	87	67.5	—
S-BI-causvid	0.33	0.40	0.144	0.232	3.15	3.36	0.673	0.016	77	73.0	-0.6 / -0.5
S-BI-rcm (consistency)	0.69	0.41	0.199	0.311	3.07	3.53	0.682	0.021	60	69.1	—
S-CAUSAL-causvid (DMD, TF)	0.03	0.56	0.075	0.120	2.56	2.55	0.678	0.018	80	73.8	-1.8 / -1.6
S-sf (DMD, SR)	0.32	0.74	0.118	0.271	3.24	3.47	0.682	0.019	62	71.6	+0.3 / +0.7
S-sf-gan	0.47	0.54	0.151	0.317	2.39	2.74	0.684	0.017	65	71.1	—
S-cf-cd (consistency)	0.75	0.53	0.192	0.464	1.35	1.81	0.679	0.011	54	63.8	—
S-rcm-tfdcm (consistency)	0.60	0.56	0.168	0.394	2.34	3.10	0.676	0.017	70	65.1	+0.7 / -1.8
S-rcm-sfdmd (CM + SR-DMD)	0.21	0.57	0.121	0.246	2.07	1.92	0.682	0.016	72	71.9	—
T-few (teacher, 4 steps)	0.35	0.50	0.169	—	4.00	8.05	0.650	0.018	49	42.5	—

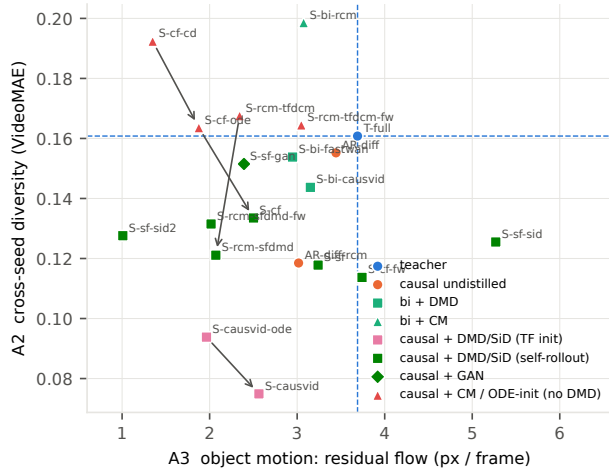


Figure 3. Every cell in the diversity–motion plane. Dashed lines: the teacher. Arrows follow public training ladders (consistency $\rightarrow$ ODE-init $\rightarrow$ DMD in Causal-Forcing; ODE-init $\rightarrow$ DMD in CausVid; consistency $\rightarrow$ self-rollout DMD in Causal-rcm): each step trades diversity for motion. Mode-seeking objectives (squares) sit low, mode-covering and adversarial ones (triangles, diamond) high.

0.192 $\rightarrow$ ODE-init 0.164 $\rightarrow$ DMD 0.133 (each step’s CI excludes zero) while object motion rises 1.35 $\rightarrow$ 1.88 $\rightarrow$ 2.50 px/frame. ODE-init is also the worst cell on every fidelity axis (recall 0.01, MUSIQ 42), so DMD’s role in these pipelines is to buy fidelity back with the diversity that remains.

Frame. Fig. 4 plots cross-seed diversity per frame. For causal DMD students the deficit is complete at frame 0 (S-causvid 0.146 vs. teacher 0.403) and the curve is flat: the KV cache propagates a collapsed first chunk and does

not deepen it, consistent with the mechanism hypothesised by [5]. Undistilled causal models show the opposite profile—diversity *grows* along the video (0.345 $\rightarrow$ 0.441) as autoregressive drift accumulates—which is precisely the temporal tax they pay (§4.4).

Chunk size—a failed prediction. If a mode-seeking objective collapses the first chunk because the chunk has few degrees of freedom, smaller chunks should collapse more. They do not: moving from 3-frame to 1-frame chunks makes Causal-Forcing’s DMD student less diverse (-0.020 [$-0.034, -0.007$]) but NVIDIA’s self-rollout DMD student *more* diverse (DINOv2 $+0.043$ [$0.012, 0.073$]), and leaves the consistency student unchanged. The first-chunk localisation stands; the degrees-of-freedom explanation does not.

4.4. Opposite signs on the other axes

Temporal coherence is refunded. Causalisation raises warp error (0.016 $\rightarrow$ 0.023; subject consistency 0.954 $\rightarrow$ 0.923; visibly, late chunks blur and drift), distillation lowers it, and the 2×2 interaction is positive in every configuration we ran—CausVid cells at CFG 3/5/7.5 ($+0.003, +0.005, +0.010$), NVIDIA cells with either student ($+0.005, +0.007$)—with the CI excluding zero in four of five (CFG 3 is borderline). Few-step students do not drift because their rollouts are short and, in the self-rollout recipes, trained on.

Camera motion is taxed more than object motion. Every student loses 40–70% of the global (camera) flow (5.7 $\rightarrow$ 1.8–3.5 px/frame) but only 10–45% of the residual (object) flow; the decomposition is needed to see that “motion collapse” is largely the loss of camera moves.

Adherence is untaxed. BLIP-2 VQA on pre-registered attribute questions is 0.66–0.69 for every cell including the

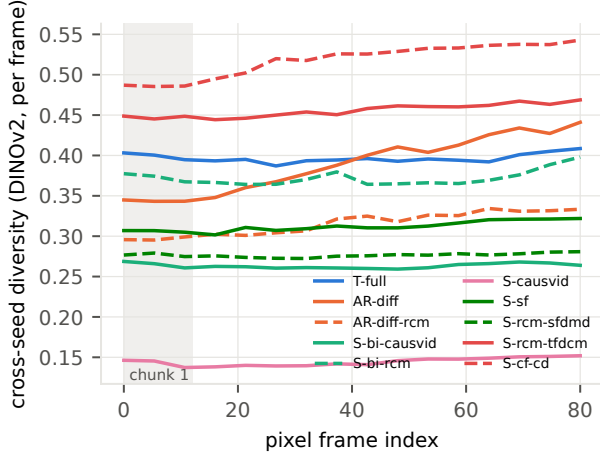


Figure 4. Per-frame cross-seed diversity (DINOv2, 8 seeds $\times$ 30 prompts). Distilled causal students are depressed from the first frame and flat; undistilled causal models drift apart along the video.

teacher; CLIP-T likewise. Mode seeking keeps each sample on-prompt; what is lost is the set, not the sample.

Oversaturation is not a DMD signature. Colourfulness rises with guidance in the teacher (CFG 3: 36, CFG 5: 50) and is 63–79 in the causal 50-step cells at matched guidance—before any distillation. DMD students (63–80) merely inherit it.

Consequence. Across 13 non-reference cells the second eigenvalue of the axis-correlation matrix carries 32% of the variance (28% among students) and 59 cell pairs (14 among students) are ordered oppositely on two axes with both differences significant. No scalar summary—VBench’s total included—can rank these models without hiding a tax.

4.5. Quality metrics are inverted on distilled video

The rater’s diversity judgements track our metrics (cell-level Spearman 0.94 with DINOv2 diversity, 0.89 with VideoMAE diversity and with recall; per-comparison agreement 83%, binomial $p < 10^{-5}$), and motion judgements track object flow at the cell level ($\rho = 0.83$). Quality is different. The rater preferred the teacher in 35.5 of 36 comparisons (BT +3.7, next best +0.7), yet MUSIQ, CLIP-IQA, MANIQA and TOPIQ—computed on eight frames per video over the whole corpus—all rank the causal DMD student *first* and the teacher fourth or lower (Fig. 5). Per-comparison agreement with the human is 47%, 36%, 39% and 36%; for CLIP-IQA and TOPIQ this is significantly below chance ($p = 0.027$). The metrics do detect blur (the 4-step teacher and the ODE-init cells score 42) but reward

the over-sharpened, over-saturated look of distilled samples. The widely reported observation that imaging quality *improves* after distillation while dynamics collapse [4] is, on this evidence, an artefact of the metric.

5. Discussion

For practitioners. Choosing a streaming student is a choice of objective before architecture. Consistency-distilled causal students keep the teacher’s coverage and diversity and pay in motion and sharpness; DMD and SiD students keep prompt adherence and per-sample fidelity and pay in coverage; adversarial training sits between. Self-rollout training and consistency initialisation each recover a quantified part of the collapse, and the ODE-regression initialisation still used by several pipelines is where most of it starts. Imaging-quality scores should not be used to compare a distilled model with its teacher.

For evaluation. The protocol is training-free and needs only public checkpoints: sample every cell on the same noise, report one primary metric per axis with paired CIs, add the undistilled causal cell whenever a causal student is evaluated, trace the teacher’s guidance curve, and calibrate any automatic quality axis against humans before reporting it.

6. Limitations

All results are on one family, Wan2.1-1.3B, the only one for which every factorial cell is public; the 14B and Wan2.2 counterparts do not fit our single 32 GB GPU. The undistilled causal cells come from two labs whose checkpoints differ in data and fine-tuning, and their main effects differ (Causal-Forcing’s model keeps diversity and drifts; NVIDIA’s loses diversity and does not); we therefore report both and draw conclusions only from contrasts that replicate. Prompts number 30 and the human study has one rater who is an author; the diversity and quality conclusions it supports should be confirmed with independent raters. All metrics are proxies; feature-space diversity is validated by the human study, temporal coherence and adherence are not. Our chunk-size prediction failed and is reported as such.

7. Conclusion

A factorial design assembled from public checkpoints shows that the diversity collapse of few-step causal video generators is an interaction between mode-seeking distillation objectives and autoregressive generation, incurred largely at initialisation and entirely within the first chunk, absent under consistency objectives, and accompanied by opposite-sign effects on temporal coherence and by quality metrics that reward the collapse. The corpus (5,405 videos), per-video metrics and all scripts are released.

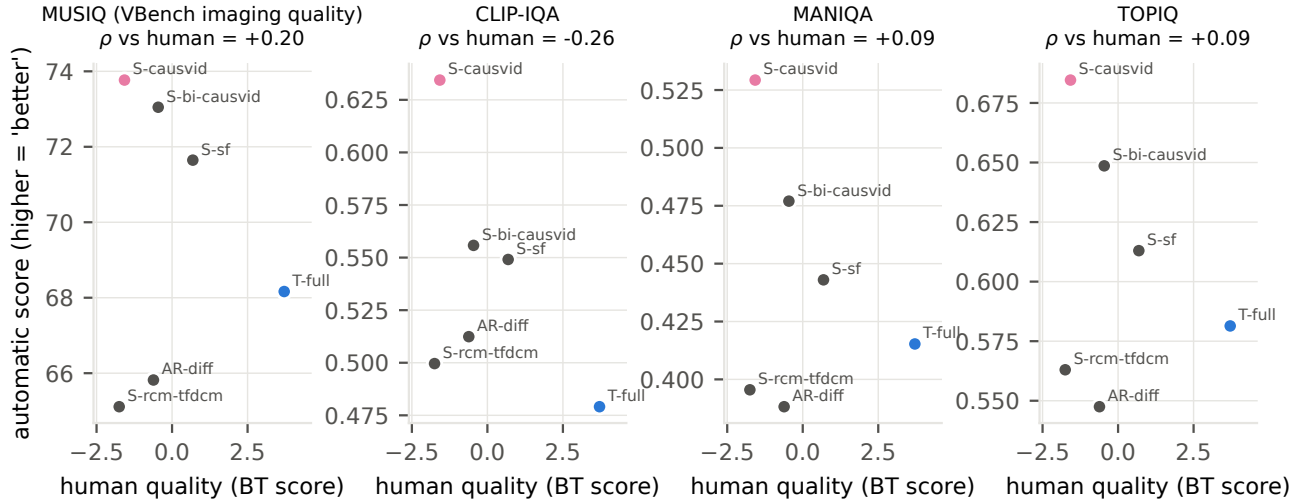


Figure 5. Four no-reference image-quality metrics against the blind human quality score (Bradley-Terry, 90 pairwise comparisons). All four place the causal DMD student (pink) above the teacher (blue); the human places the teacher above every student.

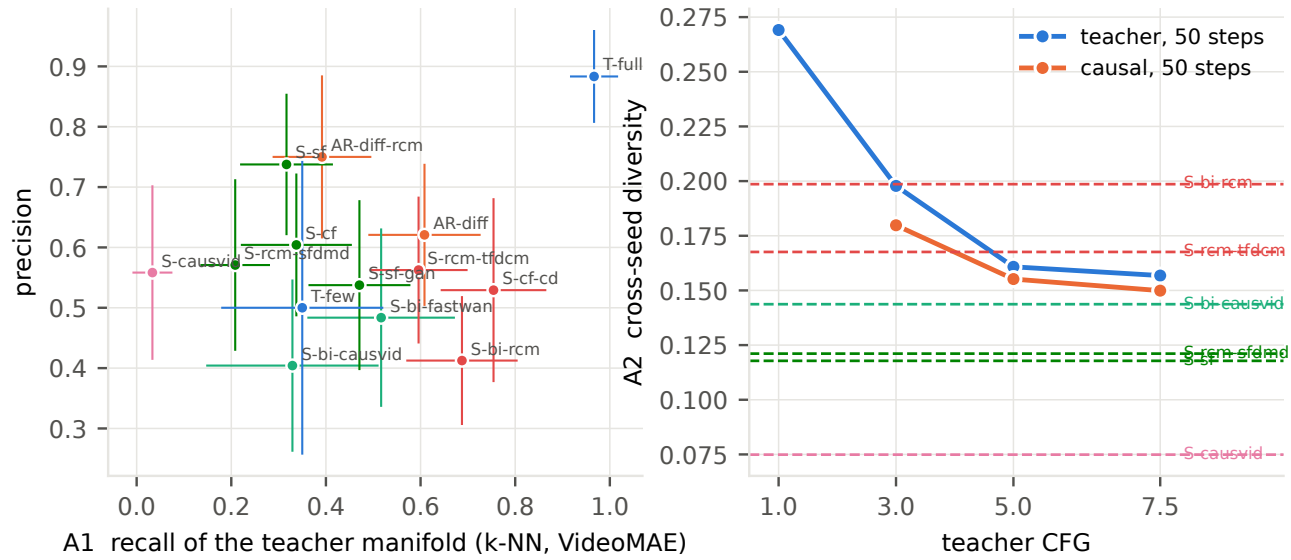


Figure 6. Left: coverage against fidelity (jackknife 95% CIs). Right: the teacher's and the undistilled causal model's diversity as a function of guidance, with students as horizontal lines; causal DMD students lie below any guidance setting.

References

- [1] Anonymous. Adaptive video distillation: Mitigating oversaturation and temporal collapse in few-step generation. *arXiv preprint arXiv:2603.21864*, 2026. 1, 2
- [2] Anonymous. Codmmd: Scale-dependent degradation in distribution matching distillation of video models. *arXiv preprint arXiv:2606.21982*, 2026. 2
- [3] Anonymous. Data-forcing distillation: Restoring diversity and fidelity in few-step video generation. *arXiv preprint arXiv:2606.18478*, 2026. 1, 2
- [4] Anonymous. Dynaforcing: Dynamic collapse in self-forcing video distillation. *arXiv preprint arXiv:2608.17707*, 2026. 1, 2, 6
- [5] Anonymous. Uncertainty dmd: Restoring diversity in few-step autoregressive video distillation. *arXiv preprint arXiv:2609.11265*, 2026. 1, 2, 5
- [6] Chaofeng Chen et al. Topiq: A top-down approach from semantics to distortions for image quality assessment. *IEEE TIP*, 2024. 2
- [7] David Hasler and Sabine Süssstrunk. Measuring colourful-

- ness in natural images. *Proc. SPIE Human Vision and Electronic Imaging*, 2003. 5
- [8] Xun Huang, Zhengqi Li, Guande He, Mingyuan Zhou, and Eli Shechtman. Self forcing: Bridging the train-test gap in autoregressive video diffusion. *arXiv preprint arXiv:2506.08009*, 2025. 1, 2
- [9] Ziqi Huang, Yinan He, Jiashuo Yu, Fan Zhang, Chenyang Si, Yuming Jiang, Yuanhan Zhang, Tianxing Wu, Qingyang Jin, Nattapol Chanpaisit, Yaohui Wang, Xinyuan Chen, Limin Wang, Dahua Lin, Yu Qiao, and Ziwei Liu. Vbench: Comprehensive benchmark suite for video generative models. In *CVPR*, 2024. 2
- [10] Junjie Ke, Qifei Wang, Yilin Wang, Peyman Milanfar, and Feng Yang. Musiq: Multi-scale image quality transformer. In *ICCV*, 2021. 2
- [11] Tuomas Kynkäänniemi, Tero Karras, Samuli Laine, Jaakko Lehtinen, and Timo Aila. Improved precision and recall metric for assessing generative models. In *NeurIPS*, 2019. 2, 3
- [12] Hao AI Lab. Fastwan: Generating a 5-second video in 5 seconds via sparse distillation. https://hao-ai-lab.github.io/blogs/fastvideo_post_training/, 2025. 2
- [13] Muhammad Ferjad Naeem, Seong Joon Oh, Youngjung Uh, Yunjey Choi, and Jaeyun Yoo. Reliable fidelity and diversity metrics for generative models. In *ICML*, 2020. 2
- [14] Maxime Oquab et al. Dinov2: Learning robust visual features without supervision. *TMLR*, 2024. 2
- [15] Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. Videomae: Masked autoencoders are data-efficient learners for self-supervised video pre-training. In *NeurIPS*, 2022. 2
- [16] Jianyi Wang, Kelvin C.K. Chan, and Chen Change Loy. Exploring clip for assessing the look and feel of images. In *AAAI*, 2023. 2
- [17] Sidi Yang et al. Maniqa: Multi-dimension attention network for no-reference image quality assessment. In *CVPR Workshops*, 2022. 2
- [18] Tianwei Yin, Michaël Gharbi, Taesung Park, Richard Zhang, Eli Shechtman, Frédo Durand, and William T. Freeman. Improved distribution matching distillation for fast image synthesis. In *NeurIPS*, 2024. 2
- [19] Tianwei Yin, Michaël Gharbi, Richard Zhang, Eli Shechtman, Frédo Durand, William T. Freeman, and Taesung Park. One-step diffusion with distribution matching distillation. In *CVPR*, 2024. 2
- [20] Tianwei Yin, Qiang Zhang, Richard Zhang, William T. Freeman, Frédo Durand, Eli Shechtman, and Xun Huang. From slow bidirectional to fast autoregressive video diffusion models. In *CVPR*, 2025. arXiv:2412.07772. 1, 2
- [21] Kaiwen Zheng et al. Large scale diffusion distillation via score-regularized continuous-time consistency. *arXiv preprint arXiv:2510.08431*, 2025. 2
- [22] Kaiwen Zheng et al. Causal-rcm: A unified teacher-forcing and self-forcing open recipe for autoregressive diffusion distillation in streaming video generation and interactive world models. *arXiv preprint arXiv:2606.25473*, 2026. 1, 2
- [23] Mingyuan Zhou, Huangjie Zheng, Zhendong Wang, Mingzhang Yin, and Hai Huang. Score identity distillation: Exponentially fast distillation of pretrained diffusion models for one-step generation. *ICML*, 2024. 2
- [24] Hongzhou Zhu et al. Causal forcing: Autoregressive diffusion distillation done right for high-quality real-time interactive video generation. In *ICML*, 2026. arXiv:2602.02214. 1, 2